Smart focus- Admin Controlled Computer Monitoring and Management System

Noorul Aatif Rahman,Jeevanandham, Hariharan

Department ofComputer Science and Engineering (Artificial Intelligence & Machine Learning)Vels Institute ofScience, Technology&AdvancedStudies (VISTAS), Chennai, India

Abstract:
Managing large numbers of connected workstations in modern educational institutions is increasingly difficult because administrators must monitor system health, application usage, user sessions, and policy compliance across distributed devices. This paper presents a distributed workstation monitoring and management system designed for college laboratories and institutional environments. The proposed architecture combines endpoint monitoring agents, secure telemetry channels, centralized reporting, and administrative dashboards to improve visibility, reduce operational effort, and support proactive response to system anomalies. The system is designed to provide real-time health indicators, application awareness, and secure remote administrative controls while preserving data integrity and accountability. Experimental evaluation and system design analysis show that the proposed framework is scalable, practical, and suitable for deployment in academic lab environments with moderate to large numbers of computers. Index Terms-workstation monitoring, distributed systems, system administration, proactive alerting, secure telemetry, academic labs.

Key words: Real-Time Monitoring,proactive Alerting,Remote Management, Academic Laboratories

Institute of Computer Science and Engineering, SIMATS Engineering, Chennai.
ISBN No.: 978-93-344-7370-4